

DEVOPS-007 — Dockerfile & Custom Docker Image

TASK 1 — Understand Dockerfile

1. What is a Dockerfile?

A Dockerfile is a text file containing instructions that Docker uses to build a Docker image. It defines the base image, application files, dependencies, and the command to run the application.

2. Why do we need a Dockerfile?

A Dockerfile automates the image-building process. It helps create consistent, repeatable, and portable Docker images for applications.

3. What is FROM?

FROM specifies the base image used to build a new Docker image. It is usually the first instruction in a Dockerfile.

Example: FROM ubuntu

4. What is WORKDIR?

WORKDIR sets the working directory inside the container. All following commands such as COPY, RUN, and CMD can operate from this directory.

Example: WORKDIR /app

5. What is COPY?

COPY copies files or directories from the Docker build context into the Docker image. It is commonly used to copy application source code and configuration files.

6. What is RUN?

RUN executes a command during the image-building process. It is commonly used to install packages, dependencies, or perform configuration.

Example: RUN apt-get update

7. What is CMD?

CMD defines the default command that runs when a container starts. It is normally used to start the application inside the container.

Example: CMD ["nginx", "-g", "daemon off;"]

8. What is EXPOSE?

EXPOSE documents the port on which the application inside the container is

expected to listen. It does not automatically publish the port to the host.

Example: EXPOSE 80

9. Difference between RUN and CMD?

RUN is executed during the Docker image build and is used to install or configure things. CMD is executed when the container starts and defines its default runtime command.

10. Difference between CMD and ENTRYPOINT?

CMD provides a default command or arguments that can easily be overridden. ENTRYPOINT defines the main executable of the container and is intended to remain the primary command. CMD is often used to provide default arguments to an ENTRYPOINT.

TASK 2 — Create Your First Dockerfile

```
akhil_chary@DESKTOP-1EMNQEB:~$ mkdir docker-day7
akhil_chary@DESKTOP-1EMNQEB:~$ cd docker-day7
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7$ touch Dockerfile README.md
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7$ ls
Dockerfile  README.md
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7$ nano Dockerfile
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7$ nano README.md
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7$ cat Dockerfile
FROM ubuntu:24.04

WORKDIR /app

COPY . .

CMD ["echo", "Hello from my Docker image"]
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7$ cat README.md
# Docker Day 7

This project demonstrates how to create a Docker image using a Dockerfile.

## Dockerfile Instructions

- FROM – selects the base image
- WORKDIR – sets the working directory
- COPY – copies project files into the image
- CMD – defines the default command
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7$
```

TASK 3 — Build Your First Image

```
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7$ sudo docker build -t devops-day7:v1 .
[sudo] password for akhil_chary:
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  3.072kB
Step 1/4 : FROM ubuntu:24.04
24.04: Pulling from library/ubuntu
966c395d29cb: Pulling fs layer
4029a2d69959: Download complete
966c395d29cb: Download complete
966c395d29cb: Pull complete
Digest: sha256:561618e2c15bf2397621dd04f96926663a3b5616c189cf7e38db7e82f5c538ea
Status: Downloaded newer image for ubuntu:24.04
---> 561618e2c15b
Step 2/4 : WORKDIR /app
---> Running in 55e8ade8ea70
---> Removed intermediate container 55e8ade8ea70
---> 031d03c3d8a2
Step 3/4 : COPY . .
---> 79cd1f48cc28
Step 4/4 : CMD ["echo", "Hello from my Docker image"]
---> Running in b43d152818b8
---> Removed intermediate container b43d152818b8
---> 7a2e5aac3322
Successfully built 7a2e5aac3322
Successfully tagged devops-day7:v1
```

```
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7$ sudo docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
devops-day7:v1	7a2e5aac3322	117MB	29.8MB	
hello-world:latest	5dd0d3e6e255	25.9kB	9.49kB	U
ubuntu:24.04	561618e2c15b	119MB	31.7MB	
ubuntu:latest	678c6550cc43	160MB	45.3MB	U

```
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7$ sudo docker run devops-day7:v1
Hello from my Docker image
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7$ sudo docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
9c34690d4c81	devops-day7:v1	"echo 'Hello from my..."	15 seconds ago	Exited (0) 14 seconds ago		confident_booth
e67f122a5a20	ubuntu	"/bin/bash"	4 hours ago	Up 2 hours		distracted_poitras
dde1d7cf34e2	ubuntu	"/bin/bash"	4 hours ago	Exited (0) 4 hours ago		vigilant_visvesvaraya
4a6415722bf8	ubuntu	"/bin/bash"	4 hours ago	Exited (0) 4 hours ago		competent_cannon
21da05c75581	ubuntu	"/bin/bash"	4 hours ago	Exited (0) 4 hours ago		strange_wright
4fa483a75d8b	hello-world	"/hello"	4 hours ago	Exited (0) 4 hours ago		awesome_hodgkin
7a7b45b8cfa4	hello-world	"/hello"	4 hours ago	Exited (0) 4 hours ago		lucid_williams

```
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7$
```

TASK 4— Understand Docker Build Context

```
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7$ sudo docker build -t devops-day7:v2 .
[sudo] password for akhil_chary:
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  3.072kB
Step 1/4 : FROM ubuntu:24.04
--> 561618e2c15b
Step 2/4 : WORKDIR /app
--> Using cache
--> 031d03c3d8a2
Step 3/4 : COPY . .
--> Using cache
--> 79cd1f48cc28
Step 4/4 : CMD ["echo", "Hello from my Docker image"]
--> Using cache
--> 7a2e5aac3322
Successfully built 7a2e5aac3322
Successfully tagged devops-day7:v2
```

```
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7$ touch secret.txt
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7$ echo "my secret data" > secret.txt
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7$ nano .dockerignore
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7$ ls -la
total 24
drwxr-xr-x  2 akhil_chary akhil_chary 4096 Aug 13 21:55 .
drwxr-x--- 15 akhil_chary akhil_chary 4096 Aug 13 03:46 ..
-rw-r--r--  1 akhil_chary akhil_chary   27 Aug 13 21:55 .dockerignore
-rw-r--r--  1 akhil_chary akhil_chary   86 Aug 13 03:47 Dockerfile
-rw-r--r--  1 akhil_chary akhil_chary  280 Aug 13 03:48 README.md
-rw-r--r--  1 akhil_chary akhil_chary   15 Aug 13 21:54 secret.txt
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7$ sudo docker build -t devops-day7:v2 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  4.096kB
Step 1/4 : FROM ubuntu:24.04
--> 561618e2c15b
Step 2/4 : WORKDIR /app
--> Using cache
--> 031d03c3d8a2
Step 3/4 : COPY . .
--> e8cdd4f5a090
Step 4/4 : CMD ["echo", "Hello from my Docker image"]
--> Running in b7d62805c95c
--> Removed intermediate container b7d62805c95c
--> 7eea980b4b5d
Successfully built 7eea980b4b5d
Successfully tagged devops-day7:v2
```

Why .dockerignore is important

`.dockerignore` prevents unnecessary or sensitive files from being sent to the Docker builder.

TASK 5 — Build a Real Static Website Image

```
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7$ mkdir website
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7$ cd website
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ touch Dockerfile index.html styles.css
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ ls
Dockerfile index.html styles.css
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ nano index.html
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ nano styles.css
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ nano Dockerfile
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker build -t devops-website:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/
```

```
Sending build context to Docker daemon 4.608kB
```

```
Step 1/5 : FROM nginx:latest
latest: Pulling from library/nginx
```

```
5a4222b844e8: Pulling fs layer
d84ae7b21412: Pulling fs layer
26c307b5e35a: Pulling fs layer
3c55dc422a81: Pulling fs layer
b8b80b9bc028: Pulling fs layer
c0df8d325117: Pulling fs layer
f5de6e85ac74: Pulling fs layer
f5de6e85ac74: Download complete
5a4222b844e8: Download complete
b8b80b9bc028: Download complete
c0df8d325117: Download complete
d84ae7b21412: Download complete
0f03cb4db0ef: Download complete
92fcf0fc2ef2: Download complete
26c307b5e35a: Download complete
26c307b5e35a: Pull complete
```

```
b8b80b9bc028: Pull complete
Digest: sha256:8541484afbc9c8a5a8a99b379568ebbc957f658583ec9448fc43104229c03cf8
Status: Downloaded newer image for nginx:latest
--> 8541484afbc9
```

```
Step 2/5 : COPY index.html /usr/share/nginx/html/index.html
--> 4cb3012e8fe8
```

```
Step 3/5 : COPY styles.css /usr/share/nginx/html/styles.css
--> efd542a6b7d8
```

```
Step 4/5 : EXPOSE 80
--> Running in a79272ddc70d
--> Removed intermediate container a79272ddc70d
--> cb86a96683b9
```

```
Step 5/5 : CMD ["nginx", "-g", "daemon off;"]
--> Running in d002379725c1
--> Removed intermediate container d002379725c1
--> 682a912d0c02
```

```
Successfully built 682a912d0c02
```

```
Successfully tagged devops-website:v1
```

```
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
devops-day7:v1	7a2e5aac3322	117MB	29.8MB	
devops-day7:v2	7eea980b4b5d	117MB	29.8MB	
devops-website:v1	682a912d0c02	238MB	63.1MB	
hello-world:latest	5dd0d3e6e255	25.9kB	9.49kB	
nginx:latest	8541484afbc9	241MB	66MB	
ubuntu:24.04	561618e2c15b	119MB	31.7MB	
ubuntu:latest	678c6550cc43	160MB	45.3MB	

```
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker run -d --name devops-web -p 8080:80 devops-website:v1
```

```
cca11295eb6d3840ced0870bef2b4d0ed5bafa60df23048634cc4cec9be7697d
```

```
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker ps
```

 Info →  In Use

Activate Windows
Go to Settings to activate Windows.

```

akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker run -d --name devops-web -p 8080:80 devops-website:v1
cca11295eb6d3840ced0870bef2b4d0ed5bafa60df23048634cc4cec9be7697d
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
cca11295eb6d	devops-website:v1	"/docker-entrypoint...."	10 seconds ago	Up 10 seconds	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
e67f122a5a20	ubuntu	"/bin/bash"	22 hours ago	Up 20 hours	

```

distracted_poitras

```

DevOps Training Portal

Name: Akhil

Role: DevOps Trainee

Training Day: Day 7

Technologies: Linux, Git, Docker, Nginx

TASK 6 — Understand Port Mapping

```

akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker port devops-web
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker inspect devops-web
[
  {
    "Id": "d818b36cf958fd0bbe67ce333d9a698d2b481ab0a56bd2a1d648dc03f18620ac",
    "Created": "2026-08-14T05:22:22.966126701Z",
    "Path": "echo",
    "Args": [
      "Hello from my Docker image"
    ],
    "State": {
      "Status": "exited",
      "Running": false,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 0,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2026-08-14T05:29:45.141691Z",
      "FinishedAt": "2026-08-14T05:29:45.39943489Z"
    },
    "Image": "sha256:7a2e5aac332213274c7021952e8d7f550008a0c644ad75f22744b0fb98ade728",
    "ResolvConfPath": "/var/lib/docker/containers/d818b36cf958fd0bbe67ce333d9a698d2b481ab0a56bd2a1d648dc03f18620ac/resolv.conf",
    "HostnamePath": "/var/lib/docker/containers/d818b36cf958fd0bbe67ce333d9a698d2b481ab0a56bd2a1d648dc03f18620ac/hostname",
    "HostsPath": "/var/lib/docker/containers/d818b36cf958fd0bbe67ce333d9a698d2b481ab0a56bd2a1d648dc03f18620ac/hosts",
    "LogPath": "/var/lib/docker/containers/d818b36cf958fd0bbe67ce333d9a698d2b481ab0a56bd2a1d648dc03f18620ac/d818b36cf958fd0bbe67ce333d9a698d2b481ab0a56bd2a1d648dc03f18620ac-json.log",
    "Name": "/devops-web",
    "RestartCount": 0,
    "Driver": "overlayfs",

```

Activate Windows
Go to Settings to activate Windows.

```

        "IPAddress": "",
        "MacAddress": "",
        "IPPrefixLen": 0,
        "IPv6Gateway": "",
        "GlobalIPv6Address": "",
        "GlobalIPv6PrefixLen": 0,
        "DNSNames": null
    }
},
"ImageManifestDescriptor": {
    "mediaType": "application/vnd.oci.image.manifest.v1+json",
    "digest": "sha256:7a2e5aac332213274c7021952e8d7f550008a0c644ad75f22744b0fb98ade728",
    "size": 965,
    "platform": {
        "architecture": "amd64",
        "os": "linux"
    }
}
}
]
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker inspect -f '{{json .HostConfig.PortBindings}}' devops-web
{"80/tcp":[{"HostIp":"","HostPort":"8080"}]}
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker stop devops-web
devops-web
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker rm devops-web
devops-web
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker run -d --name devops-web -p 8081:80 devops-website:v1
113accc477e567f4d43d50a51ef890e694bcdda3147a8f4fd4ce6cb50d73f7f6

```

```

akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
113accc477e5   devops-website:v1  "/docker-entrypoint..."  10 seconds ago Up 10 seconds  0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
devops-web
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker port devops-web
80/tcp -> 0.0.0.0:8081
80/tcp -> [::]:8081

```

DevOps Training Portal

Name: Akhil

Role: DevOps Trainee

Training Day: Day 7

Technologies: Linux, Git, Docker, Nginx

TASK 7 — Docker Image Versioning

```
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker build -t devops-website:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  4.608kB
Step 1/5 : FROM nginx:latest
--> 8541484afbc9
Step 2/5 : COPY index.html /usr/share/nginx/html/index.html
--> Using cache
--> 4cb3012e8fe8
Step 3/5 : COPY styles.css /usr/share/nginx/html/styles.css
--> Using cache
--> efd542a6b7d8
Step 4/5 : EXPOSE 80
--> Using cache
--> cb86a96683b9
Step 5/5 : CMD ["nginx", "-g", "daemon off;"]
--> Using cache
--> 682a912d0c02
Successfully built 682a912d0c02
Successfully tagged devops-website:v1
```

```
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ nano index.html
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker build -t devops-website:v2 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  4.608kB
Step 1/5 : FROM nginx:latest
--> 8541484afbc9
Step 2/5 : COPY index.html /usr/share/nginx/html/index.html
--> f53eb1e9160d
Step 3/5 : COPY styles.css /usr/share/nginx/html/styles.css
--> 3d41099e02d2
Step 4/5 : EXPOSE 80
--> Running in db01248465b7
--> Removed intermediate container db01248465b7
--> b7f24d287680
Step 5/5 : CMD ["nginx", "-g", "daemon off;"]
--> Running in 89f6e87c2c0f
--> Removed intermediate container 89f6e87c2c0f
--> bb150db6eca8
Successfully built bb150db6eca8
Successfully tagged devops-website:v2
```

```
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker build -t devops-website:latest .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  4.608kB
Step 1/5 : FROM nginx:latest
--> 8541484afbc9
Step 2/5 : COPY index.html /usr/share/nginx/html/index.html
--> Using cache
--> f53eb1e9160d
Step 3/5 : COPY styles.css /usr/share/nginx/html/styles.css
--> Using cache
--> 3d41099e02d2
Step 4/5 : EXPOSE 80
--> Using cache
--> b7f24d287680
Step 5/5 : CMD ["nginx", "-g", "daemon off;"]
--> Using cache
--> bb150db6eca8
Successfully built bb150db6eca8
Successfully tagged devops-website:latest
```



```
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
devops-day7:v1	7a2e5aac3322	117MB	29.8MB	U
devops-day7:v2	7eea980b4b5d	117MB	29.8MB	
devops-website:latest	bb150db6eca8	238MB	63.1MB	
devops-website:v1	682a912d0c02	238MB	63.1MB	U
devops-website:v2	bb150db6eca8	238MB	63.1MB	
hello-world:latest	5dd0d3e6e255	25.9kB	9.49kB	U
nginx:latest	8541484afbc9	241MB	66MB	
ubuntu:24.04	561618e2c15b	119MB	31.7MB	
ubuntu:latest	678c6550cc43	160MB	45.3MB	U

```
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$
```

Companies should use meaningful image tags instead of relying only on latest because tags such as v1, v2, or 1.0 clearly identify a specific version of an application. This makes deployments reproducible, simplifies troubleshooting and rollback, and helps teams know exactly which image version is running in each environment. The latest tag can change over time, so using it alone can create unpredictable deployments.

TASK 8 — Dockerfile Troubleshooting

```
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website\$ ls				
Dockerfile	dockerfile	index.html	styles.css	

```
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo vim Dockerfile
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker build -t devops-nginx-new .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
             Install the buildx component to build images with BuildKit:
             https://docs.docker.com/go/buildx/

Sending build context to Docker daemon   5.12kB
Step 1/5 : FROM nginx:latest
latest: Pulling from library/nginx
5a4222b844e8: Pulling fs layer
d84ae7b21412: Pulling fs layer
26c307b5e35a: Pulling fs layer
3c55dc422a81: Pulling fs layer
b8b80b9bc028: Pulling fs layer
c0df8d325117: Pulling fs layer
f5de6e85ac74: Pulling fs layer
5a4222b844e8: Download complete
f5de6e85ac74: Download complete
b8b80b9bc028: Download complete
d84ae7b21412: Download complete
c0df8d325117: Download complete
0f03cb4db0ef: Download complete
92fcf0fc2ef2: Download complete
3c55dc422a81: Download complete
26c307b5e35a: Download complete
26c307b5e35a: Pull complete
```

```

akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
devops-nginx-new:latest	80d55e8f0b9d	238MB	63.1MB	
nginx:latest	8541484afbc9	241MB	66MB	

```

akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker run -d --name devops-nginx -p 8080:8080 devops-nginx-new:latest
6c183586c8649fc7cd15c73303e6a9015a0de9b6632d39f546ba3807f68f5ba3
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ docker ps
permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
6c183586c864	devops-nginx-new:latest	"/docker-entrypoint..."	17 seconds ago	Up 16 seconds	80/tcp, 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp

```

akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ ^C
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker logs devops-nginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/08/14 09:28:42 [notice] 1#1: using the "epoll" event method
2026/08/14 09:28:42 [notice] 1#1: nginx/1.31.3
2026/08/14 09:28:42 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/08/14 09:28:42 [notice] 1#1: OS: Linux 6.18.33.2-microsoft-standard-WSL2
2026/08/14 09:28:42 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:1048576
2026/08/14 09:28:42 [notice] 1#1: start worker processes
2026/08/14 09:28:42 [notice] 1#1: start worker process 29

```

```

akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker inspect devops-nginx
[
  {
    "Id": "6c183586c8649fc7cd15c73303e6a9015a0de9b6632d39f546ba3807f68f5ba3",
    "Created": "2026-08-14T09:28:41.851201415Z",
    "Path": "/docker-entrypoint.sh",
    "Args": [
      "nginx",
      "-g",
      "daemon off;"
    ],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 17282,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2026-08-14T09:28:41.964335375Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    },
    "Image": "sha256:80d55e8f0b9d679fa7a2db0e6e87bff16e13229e1024d2615c122de5235c512e",
    "ResolvConfPath": "/var/lib/docker/containers/6c183586c8649fc7cd15c73303e6a9015a0de9b6632d39f546ba3807f68f5ba3/resolv.conf",
    "HostnamePath": "/var/lib/docker/containers/6c183586c8649fc7cd15c73303e6a9015a0de9b6632d39f546ba3807f68f5ba3/hostname",
    "HostsPath": "/var/lib/docker/containers/6c183586c8649fc7cd15c73303e6a9015a0de9b6632d39f546ba3807f68f5ba3/hosts",
    "LogPath": "/var/lib/docker/containers/6c183586c8649fc7cd15c73303e6a9015a0de9b6632d39f546ba3807f68f5ba3/6c183586c8649fc7cd15c73303e6a9015a0de9b6632d39f546ba3807f68f5ba3-json.log"
  }
]

```

```

    "ImageManifestDescriptor": {
      "mediatype": "application/vnd.oci.image.manifest.v1+json",
      "digest": "sha256:80d55e8f0b9d679fa7a2db0e6e87bff16e13229e1024d2615c122de5235c512e",
      "size": 2263,
      "platform": {
        "architecture": "amd64",
        "os": "linux"
      }
    }
  }
}

```

```

akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ curl http://localhost:8080
curl: (56) Recv failure: Connection reset by peer
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker run -d --name devops-nginx1 -p 80:8080 devops-nginx-new:latest
e16d406cb65d097ae0c3291942f4cc7666fdeac19caf87768815ffc4dbd20a6
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ curl http://localhost:8080
curl: (56) Recv failure: Connection reset by peer
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ ^C
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker rm -f devops-nginx1
devops-nginx1
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker rm -f devops-nginx
devops-nginx
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker run -d --name devops-nginx1 -p 80:8080 devops-nginx-new:latest
b35a65d12526a1e2a2a9bd2c9d20d95fba6d21006e7074c4f42a190aa27d9fc6
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker rm -f devops-nginx1
devops-nginx1
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  STATUS   PORTS   NAMES
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker run -d --name devops-nginx -p 8080:80 devops-nginx-new:latest
54a67a92befc18ce28c01adb3315d05c050457296e719a5ed68480eb1fa75019
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker ps -a

```

```

akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker run -d --name devops-nginx -p 8080:80 devops-nginx-new:latest
54a67a92befc18ce28c01adb3315d05c050457296e719a5ed68480eb1fa75019
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
54a67a92befc   devops-nginx-new:latest             "/docker-entrypoint. ..." 11 seconds ago Up 11 seconds 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   devops-nginx
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ curl http://localhost:8080
<h1>Docker Day 7 - Troubleshooting</h1>

```

← → ↻ ⓘ localhost:8080

Docker Day 7 - Troubleshooting

TASK 9 — Push Image to Docker Hub

```

akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker images
[sudo] password for akhil_chary:
IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
devops-nginx-new:latest              80d55e8f0b9d                    238MB       63.1MB       u
nginx:latest                         8541484afbc9                    241MB       66MB
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker login

USING WEB-BASED LOGIN

Info → To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: MZKD-LNLD
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...
^Clogin canceled
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker login -u akhilblackroth

Info → A Personal Access Token (PAT) can be used instead.
       To create a PAT, visit https://app.docker.com/settings

Password:
Error response from daemon: Get "https://registry-1.docker.io/v2/": unauthorized: incorrect username or password

```

Activate Windows
Go to Settings to activate Windows.

```
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker login -u akhilblackroth

Info → A Personal Access Token (PAT) can be used instead.
To create a PAT, visit https://app.docker.com/settings

Password:
Error response from daemon: Get "https://registry-1.docker.io/v2/": unauthorized: incorrect username or password
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker login -u akhilblackroth

Info → A Personal Access Token (PAT) can be used instead.
To create a PAT, visit https://app.docker.com/settings

Password:

WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker tag devops-nginx-new:latest akhil123/devops-day7:v1
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker images
```

```
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker tag devops-nginx-new:latest akhilblackroth/devops-day7:v1
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker images

Info → U In Use

IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
akhil123/devops-day7:v1              80d55e8f0b9d                     238MB        63.1MB        U
akhilblackroth/devops-day7:v1        80d55e8f0b9d                     238MB        63.1MB        U
devops-nginx-new:latest              80d55e8f0b9d                     238MB        63.1MB        U
nginx:latest                         8541484afbc9                     241MB        66MB          U
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$ sudo docker push akhilblackroth/devops-day7:v1
The push refers to repository [docker.io/akhilblackroth/devops-day7]
f5de6e85ac74: Mounted from library/nginx
b8b80b9bc028: Mounted from library/nginx
255709990694: Pushed
26c307b5e35a: Mounted from library/nginx
3c55dc422a81: Mounted from library/nginx
d939c8c5f277: Pushed
c0df8d325117: Mounted from library/nginx
5a4222b844e8: Mounted from library/nginx
d84ae7b21412: Mounted from library/nginx
v1: digest: sha256:80d55e8f0b9d679fa7a2db0e6e87bffa16e13229e1024d2615c122de5235c512e size: 2263
akhil_chary@DESKTOP-1EMNQEB:~/docker-day7/website$
```

akhilblackroth/devops-day7

Last pushed 1 minute ago · Repository size: 60.2 MB · ☆0 · ↓0

Add a description

Add a category

Docker commands

To push a new tag to this repository:

docker push akhilblackroth/devops-day7:tagname

General

Tags

Image Management

Collaborators

Webhooks

Settings

Sort by

Newest

Filter tags

Delete

TAG

v1

Last pushed 1 minute by akhilblackroth

docker pull akhilblackroth/devops-day7:v1

Digest	OS/ARCH	Last pull	Compressed size
80d55e8f0b9d	linux/amd64	less than 1 day	60.19 MB

Activate Windows
Go to Settings to activate Windows.

TASK 10 — Final Practical Challenge

[Repositories](#) / [devops-day7](#) / [Tags](#)

akhilblackroth/devops-day7 

Last pushed 1 minute ago · Repository size: 60.2 MB · ☆ 0 · ↓ 0

Add a description  

Add a category  

Using 0 of 1 private repositories.

Docker commands

To push a new tag to this repository:

[Public view](#)

```
docker push akhilblackroth/devops-day7:tagname
```

General **Tags** Image Management Collaborators Webhooks Settings 

☐ Sort by **Newest** 

TAG  v1 Last pushed 1 minute by akhilblackroth				<pre>docker pull akhilblackroth/devops-day7:v1</pre> 	
☐ Digest		OS/ARCH	Last pull	Compressed size 	
80d55e8f0b9d		linux/amd64	less than 1 day	60.19 MB	

Activate Windows
Go to Settings to activate Windows.

